MIL-P-36627A
16 August 1977
SUPERSEDING
MIL-P-36627 dated
27 February 1970 and
MIL-P-36626 dated
16 April 1970

MILITARY SPECIFICATION

PIN, BONE, STEINMANN, BAYONET AND TROCAR POINT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bone pins of the Steinmann design, bayonet and trocar point, for traction reduction of fractures.

1.2 Classification. Bone pins shall be of the following types, sizes and styles as specified (see 6.1):

Type I - Bayonet point.
Type II - Trocar point.

Size	Length (inches)
1	4
2	5
3	6-1/2
4	8
5	9
6	12

Style	Diameter (inches)
1	0.093
2	0.125
3	0.156

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-I-526 - **Instruments, Dental and Surgical; General Specifications.**
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case, and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center (DPSC)

2202 Pin, Bone, Bayonet Point, Steinmann.
1907 Pin, Bone, Trocar Point, Steinmann.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 8 Tension Testing of Metallic Materials.

ASTM E 18 Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

ASTM E 45 Determining the Inclusion Content of Steel.

ASTM F 86 Recommended Practice for Surface Preparation and Marking of Metallic Surgical Implants.

ASTM E 92 Method of Test for Vickers Hardness of Metallic Materials.

ASTM E 112 Estimating the Average Grain Size of Metals.

ASTM E 140 Standard Hardness Conversion Tables for Metals (Relationship between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

ASTM F 138 Stainless Steel Bars and Wire for Surgical Implants (Special Quality).

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC. (USP)

The Pharmacopeia of the United States of America

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Material. The bone pins shall be fabricated from cold drawn wire, conforming to the composition specified in ASTM F 138, Grade 2 (0.03 maximum carbon content). Tensile strength of wire shall be 150,000 to 200,000 pounds per square inch when tested as specified in 4.4.2.

* 3.1.1 Grain size. Grain size of the pins shall be determined and shall be 5 or finer in accordance with ASTM F 138 as determined by ASTM E 112.

* 3.1.2 Microcleanliness of material. The microcleanliness of the raw material (steel billet) shall be in accordance with ASTM F 138.

3.2 Hardness. Rockwell hardness of the bone pins shall be C33 to C43 when tested as specified in 4.4.3.

3.3 Finish. The completed bone pin shall have a bright, smooth, mirror finish, properly electropolished. Definitions of acceptable surface characteristics shall be as described in ASTM F 86.

3.3.1 Passivation. The bone pins shall be properly passivated after completion of all final finishing operations and shall be accomplished by the methods described in ASTM F 86. Under no circumstances shall electropolishing be considered to satisfy this passivation requirement.

* 3.4 Style design and dimensions. Shall be a one-piece bone pin with Type I (bayonet point) or Type II (trocar point) and triangular shank end. Shall be of the Steinmann design, suitable for implanting into the body for traction reduction of fractures. The bone pins shall be in accordance with Defense Personnel Support Center (DPSC) Drawing No. 2202, for Type I, bone pins and DPSC Drawing No. 1907, for Type II pins. Bone pins shall be furnished in accordance with the appropriate type, size, style, stock number, and item identification.

3.5 Resistance to boil test, copper sulfate test and saline test. Bone pins shall not show evidence of corrosion when subjected to the boil test as specified in 4.4.1.1, the copper sulfate test as specified in 4.4.1.2, and the saline test as specified in 4.4.1.3.

3.6 Workmanship. Workmanship shall be first class throughout. Bone pins shall be free from pits, cracks, slivers, scale, corrosion, tool marks, dull points and other defects which detract from their appearance or impair their serviceability, proper functioning and intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the cold drawn corrosion-resistant steel wire may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one bone pin.

TABLE I. Sampling criteria for examination.

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	None defined.	

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for testing purposes shall be one bone pin. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling criteria for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Bone pin	Tensile strength	3.1	4.4.2	S-2	1.5
	Grain size	3.1.1 & ASTM F138	ASTM E112	---	---
	Microcleanliness	3.1.2 & ASTM F138	ASTM E45	S-1 1/	1.0
	Hardness	3.2	4.4.5	S-2	1.5
	Dimensions	3.4	-----	S-2	1.5
	Boil test	3.5	4.4.1.1	S-4	0.65
	Copper sulfate test	3.5	4.4.1.2	S-4	0.65
	Saline test	3.5	4.4.1.3	S-4	0.65

1/ The unit lot of steel, for which the microcleanliness (inclusion content) is to be determined, shall not be larger than one heat.

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the manufacture of the bone pins shall be selected for chemical analysis.

4.3 Examination. The bone pins shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any bone pins in the sample containing one or more defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

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TABLE III. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Bone pins not of finish specified.
102	Bone pin does not have sharp bayonet point or sharp trocar point as applicable.
103	Bone pins not free from pits, cracks, slivers, scale, corrosion or tool marks.
104	Bone pin does not have triangular shank at proximal end.

4.3.1 Packaging inspection. The inspection of the preservation-packaging shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and its marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1.1 Boil test. Bone pins shall be subjected to the boil test as described and specified in GG-I-526.

4.4.1.2 Copper sulfate test. Bone pins shall be subjected to the copper sulfate test as described and specified in GG-I-526.

4.4.1.3 Saline test. Bone pins shall be scrubbed with soap and warm water, rinsed in hot water, followed by immersion into ethyl alcohol (95 percent) and dried. Bone pins shall then be completely immersed into Normal Saline Solution (USP) and while immersed, placed into an oven at a temperature of 37 ± 1 degrees Centigrade for a period of 7 days, removed from the oven and examined for corrosion while still immersed.

4.4.2 Tensile strength test. Tensile strength of the bone pin shall be tested in accordance with ASTM E 8.

4.4.3 Hardness test. Rockwell hardness of the bone pins shall be tested in accordance with ASTM E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Std. Hardness Conversion Tables E 140.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

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5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each pin shall be thoroughly cleaned and dried in accordance with the applicable methods outlined in MIL-P-116.

5.1.1.2 Preservative. Preservative shall not be utilized.

5.1.1.3 Unit package. Each pin shall be wrapped in neutral tissue or other suitable anticorrosive material. Each pin shall be further packed in accordance with MIL-P-116, method IC-1 or 1A-8.

5.1.1.4 Intermediate package. Twelve (12) pins shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676, or PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. Pins shall be clean and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Five hundred seventy six (576) pins shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix for the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure an sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Five hundred seventy six (576) pins shall be packed in a exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of PPP-B-636.

5.2.3 Level C. The packaged pins shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

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5.3 Marking. Each unit, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.
(b) Type, size and style required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Applicable National Stock Numbers.

6.3 National Stock number coverage. This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification	Specification Designation		
		Type	Size	Style
6515-00-299-8315	PIN, BONE, Bayonet Point, Steinmann, 4 inch	I	1	1
6515-00-299-8316	PIN, BONE, Bayonet Point, Steinmann, 5 inch	I	2	2
6515-00-299-8317	PIN, BONE, Bayonet Point Steinmann, 6-1/2 inch	I	3	2
6515-00-299-8318	PIN, BONE, Bayonet Point, Steinmann, 8 inch	I	4	3
6515-00-299-8319	PIN, BONE, Bayonet Point, Steinmann, 9 inch	I	5	3
6515-00-310-7880	PIN, BONE, Trocar Point, Steinmann, 4 inch	II	1	1
6515-00-310-7900	PIN, BONE, Trocar Point, Steinmann, 5 inch	II	2	2
6515-00-310-7920	PIN, BONE, Trocar Point, Steinmann, 6-1/2 inch	II	3	2
6515-00-310-7940	PIN, BONE, Trocar Point, Steinmann, 8 inch	II	4	3
6515-00-310-7960	PIN, BONE, Trocar Point, Steinmann, 9 inch	II	5	3
6515-00-664-7854	PIN, BONE, Trocar Point, Steinmann, 9 inch	II	5	1
6515-00-664-7855	PIN, BONE, Trocar Point, Steinmann, 12 inch	II	6	3

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* 6.4 Changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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DOCUMENT IDENTIFIER AND TITLE

MIL-P-36627A PIN, BONE, STEINMANN, BAYONET AND TROCAR POINT

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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